Adam Shebani

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EDUCATION

University of Wisconsin-Madison, College of Engineering

Degree: Bachelor of Science, Computer Engineering - GPA: 3.9/4.0 *Expected:* December 2027

ENGINEERING EXPERIENCE

Silicon Badgers – RTL Team Designer

- Profiled Qwen3.5-2B inference in llama.cpp (C/C++) by compute-graph operation, attributing ~90% of runtime to matrix multiplication and defining the accelerator's target workload.
- Designed a parameterizable INT4/INT8 matrix-multiplication accelerator in SystemVerilog targeting the model's attention and MLP layers.

Big Data Systems (Python) – CS 544 UW-Madison

- Built distributed storage and streaming pipelines with HDFS, Spark, Cassandra, and Kafka, deployed across Docker containers and connected via gRPC.
- Developed PySpark workflows over Parquet data, and designed relational schemas and RESTful backend services for distributed applications.

Embedded Systems and Microcontrollers (C) – ECE 353 UW-Madison

- Developed embedded C firmware on an ARM Cortex-M (PSoC 6) using interrupt-driven design with timers, GPIO, PWM, and ADC peripherals.
- Built concurrent I/O functionality using FreeRTOS to coordinate sampling, communication, and display updates without polling.

PERSONAL PROJECTS

Trading Dashboard (Python, Interactive Brokers API)

- Built a read-only market-data layer on the Interactive Brokers TWS API, wrapping its callback-driven threading in blocking calls with typed errors and tagging each quote as live, delayed, or frozen to surface data-entitlement gaps.
- Designed a local SQLite cache for historical prices that fetches only missing date ranges, validating and storing ~5,000 daily bars so repeat queries need no network access.

ML Cookbook (Python, scikit-learn, pandas)

- Built a Python library for classical ML on tabular data that infers each column's statistical type, validates analysis preconditions before running, and supports user type overrides.
- Implemented correlation, regression, decision tree, and random forest workflows returning immutable, JSON-serializable results, decoupling the core from any frontend.

Wearable Motion Tracker (Infineon PSoC 6, C, FreeRTOS)

- Brought up six peripherals over SPI, I2C, and PDM under FreeRTOS on an ARM Cortex-M4: IMU, magnetometer, pressure sensor, 60 GHz radar, MEMS microphone, and QSPI flash.
- Logged and streamed multi-sensor data to train a CNN gesture classifier, deployed on-device at 84% accuracy with 200ms inference latency.

SKILLS

- **Languages:** Python, C, C++, Java, SQL, SystemVerilog
- **ML & Data:** scikit-learn, pandas, NumPy, PySpark, llama.cpp, LLM inference profiling
- **Distributed Systems:** Spark, Kafka, Cassandra, HDFS, gRPC, REST
- **Databases:** SQLite, relational schema design
- **Tools:** Docker, Git, GitHub, Linux, Claude Code, Codex